# DISCRETE LAGUERRE INEQUALITIES FOR A COLORED PARTITION FAMILY

1200 AND GPT ASTRA (AI CONTRIBUTOR)

ABSTRACT. For integers $d \geq 2$ and $m \geq 2$, let $p_{d,m}(n)$ be the coefficient of $(1-z)^{-2d}(1-z^m)^{-1}$. We determine exactly when all discrete Laguerre inequalities of orders at most $d$ for this sequence hold strictly for all sufficiently large indices. This occurs for every $m$ when $d$ is even, and precisely for $m \leq 1 + \frac{1}{2}\binom{2d}{d}$ when $d$ is odd. Above the latter threshold, the order-$d$ transform is eventually negative on a specified residue class. At equality, its leading term vanishes on that class; we compute a positive term two degrees lower. The argument gives a cyclic-difference formula for the boundary coefficient of the order-$d$ transform of exact quasi-polynomials whose coefficients above degree $\ell - 2d$ are constant. Rational residue polynomials provide explicit, parameter-dependent thresholds and independently checkable sign certificates.

## 1. INTRODUCTION

For a real sequence $f = (f(n))_{n\geq 0}$, define its discrete Laguerre transform of order $d \geq 1$ by

$$\mathcal{T}_d(f)(n) = \sum_{j=0}^{2d} (-1)^{j+d} \binom{2d}{j} f(n+j) f(n+2d-j). \tag{1}$$

The corresponding inequality is $\mathcal{T}_d(f)(n) \geq 0$. In order one this is twice the usual adjacent log-concavity difference, with a shift of the index. We distinguish non-strict inequalities from eventual strict positivity.

Gajdzica proved sufficient conditions under which quasi-polynomial-like functions satisfy discrete Laguerre inequalities [1, Theorems 4.3–4.4]. His criterion requires the relevant upper coefficients to be constant. Examples 4.5–4.6 of that article demonstrate that this condition is not necessary; the latter example treats a multiset with four ones and one part of size 300. We retain its eventual-positivity conclusion and correct its exceptional residue and numerical leading coefficient in Section 4.1. Problem 5.6 asks for more effective criteria. The present paper studies the first periodic coefficient and gives a complete eventual classification for the family

$$\sum_{n\geq 0} p_{d,m}(n) z^n = \frac{1}{(1-z)^{2d}(1-z^m)}. \tag{2}$$

Here the $2d$ parts of size one have distinct colors. The smooth asymptotic coefficient from the earlier criterion and the isolated example just mentioned are existing results. Our principal contribution is the all-parameter classification, including the odd critical case.

**Theorem 1.1.** *Let $d \geq 2$ and $m \geq 2$ be integers. There is an explicitly computable integer $N(d,m)$ such that*

$$\mathcal{T}_j(p_{d,m})(n) > 0 \qquad (1 \leq j \leq d,\ n \geq N(d,m))$$

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*if and only if either $d$ is even, or $d$ is odd and*

$$m \le 1 + \frac{1}{2}\binom{2d}{d}. \tag{3}$$

*If $d$ is odd and (3) fails, then $\mathcal{T}_d(p_{d,m})(n) < 0$ for every sufficiently large integer $n$ satisfying $m \mid n + 2d$. Explicit thresholds are supplied by the finite coefficient formulas in Sections 3 and 4.*

The bound depends on the fixed parameters $d, m$; no bound independent of both parameters is asserted. A few consequences illustrate the parity distinction:

| Order $d$ | Allowed $m$ for eventual strict positivity | Critical value |
|---|---|---|
| 2 | every $m \ge 2$ | none |
| 3 | $2 \le m \le 11$ | 11 |
| 4 | every $m \ge 2$ | none |
| 5 | $2 \le m \le 127$ | 127 |

These entries are consequences of the theorem, rather than a finite-search claim.

We work with exact quasi-polynomials. This is a specified subclass of the broader functions in the source problem: an arbitrary unspecified little-$o$ remainder does not provide an explicit error modulus. We do not claim a classification for all restricted partition multisets or for arbitrary periodic upper coefficients.

## 2. The first periodic coefficient

A quasi-polynomial of period $M \ge 1$ is specified by polynomials $f_0(X), \dots, f_{M-1}(X)$, with $f(n) = f_r(n)$ when $n \equiv r \pmod M$. For a periodic vector $b$, all its arguments are understood modulo $M$. Put

$$\Delta b(t) = 2b(t) - b(t-1) - b(t+1).$$

It is useful to center (1):

$$\mathcal{H}_d(f)(t) = \sum_{j=-d}^{d} w_j f(t+j) f(t-j), \qquad w_j = (-1)^j \binom{2d}{d+j}.$$

Then $\mathcal{T}_d(f)(n) = \mathcal{H}_d(f)(n+d)$, and $w_j = w_{-j}$.

**Lemma 2.1.** *For the preceding weights,*

$$\sum_{j=-d}^{d} w_j j^k = 0 \quad (0 \le k < 2d), \qquad \sum_{j=-d}^{d} w_j j^{2d} = (-1)^d (2d)!.$$

*Moreover, $\sum_j w_j b(t+j) = \Delta^d b(t)$.*

*Proof.* The finite binomial identities

$$\sum_j w_j z^j = (2 - z - z^{-1})^d, \qquad \sum_j w_j e^{jv} = (2 - 2\cosh v)^d$$

give the cyclic-shift formula and the required moments. Indeed, the second expression has expansion $(-1)^d v^{2d} + O(v^{2d+2})$ as a formal power series. □

**Theorem 2.2.** *Let $d \ge 1$, $\ell \ge 2d$, and let $P$ be a real polynomial of degree $\ell$ and leading coefficient $a > 0$. Suppose*

$$f_r(X) = P(X) + b_r X^{\ell - 2d} + Q_r(X), \qquad \deg Q_r \le \ell - 2d - 1.$$

*When $\ell = 2d$, the last condition means $Q_r = 0$. The residue polynomial of $\mathcal{T}_d(f)$ has degree at most $2\ell - 2d$; its coefficient at this degree on residue $r$ is*

$$\gamma_r = (2d)!\binom{\ell}{d}a^2 + 2a\Delta^d b(r+d). \tag{4}$$

*For $1 \le j < d$, the residue polynomials of $\mathcal{T}_j(f)$ have degree $2\ell-2j$ and positive, residue-independent leading coefficient $(2j)!\binom{\ell}{j}a^2$.*

*Proof.* The coefficient of $t^k$ in $P(t+j)P(t-j)$ has degree at most $2\ell - k$ as a polynomial in $j$. Lemma 2.1 therefore annihilates the terms with $k > 2\ell - 2d$. At equality, any term involving a lower-degree monomial of $P$ still has $j$-degree less than $2d$. The leading monomials give

$$a^2(t+j)^\ell(t-j)^\ell = a^2(t^2-j^2)^\ell.$$

The coefficient of $t^{2\ell-2d}$ before weighted summation is $a^2(-1)^d\binom{\ell}{d}j^{2d}$. This yields the first term of (4).

Write $R = f - P$. On a fixed residue of $t$, the leading term of $R(t+j)$ is $b(t+j)t^{\ell-2d}$. The cross terms contribute

$$a\sum_j w_j(b(t+j) + b(t-j)) = 2a\Delta^d b(t)$$

at degree $2\ell - 2d$. Products of two $R$ terms have degree at most $2\ell - 4d$, and terms involving $Q_r$ contribute below $2\ell - 2d$. The translation $t = n + d$ preserves the highest possible coefficient and gives the residue shift in (4). For $j < d$, the periodic term is below the boundary degree for order $j$; the same argument gives the last assertion. □

If all $\gamma_r$ are positive, the order-$d$ transform is eventually strictly positive. A negative $\gamma_r$ gives infinitely many failures. A zero $\gamma_r$ supplies no conclusion without lower coefficients. The latter distinction is essential in the proof of Theorem 1.1.

## 3. Explicit bounds and residue certificates

For a nonzero polynomial $h(X) = c_kX^k + \cdots + c_0$, define

$$B(h) = 1 + \left\lceil \frac{\sum_{i=0}^{k-1}|c_i|}{|c_k|} \right\rceil. \tag{5}$$

The sum is empty for a constant polynomial.

**Lemma 3.1.** *For every real $X \ge B(h)$, $h(X)$ has the sign of $c_k$.*

*Proof.* The constant case is immediate. Otherwise put $S = \sum_{i<k}|c_i|$. Since $X \ge 1$, the absolute value of the lower-degree part is at most $SX^{k-1}$. But $X > S/|c_k|$, so the leading monomial strictly dominates. □

For exact rational input, the transform on residue $r$ is the explicitly known rational polynomial

$$h_{d,r}(X) = \sum_{j=0}^{2d}(-1)^{j+d}\binom{2d}{j}f_{r+j}(X+j)f_{r+2d-j}(X+2d-j). \tag{6}$$

The residue subscripts are reduced modulo $M$. A zero polynomial has sign zero throughout that residue; every nonzero polynomial has the eventual sign of its leading nonzero coefficient. For one order a common bound is

$$N = \max(\{1\} \cup \{B(h_{d,r}) : h_{d,r} \ne 0\}).$$

For orders $1, \dots, d$, take the same maximum over the order as well. This convention gives $N = 1$ even if all output polynomials vanish. Zero polynomials never count as strictly positive.

Thus an exact certificate consists of input coefficient vectors, the vectors in (6), their actual leading signs, and the integer bounds (5). Rational arithmetic checks all identities and inequalities. This elementary polynomial-sign mechanism is used for effectivity; it is not itself presented as a new general principle.

## 4. Exact decomposition of the partition family

Fix $q = 2d$ and $m \geq 2$. From (2),

$$p_{d,m}(n) = \sum_{v=0}^{\lfloor n/m \rfloor} \binom{n - mv + q - 1}{q - 1}. \tag{7}$$

Set $u_0 = 1/m$, and recursively define

$$u_h = -\frac{1}{m} \sum_{k=1}^{\min(h,m-1)} (-1)^k \binom{m}{k+1} u_{h-k} \qquad (1 \leq h \leq q). \tag{8}$$

Let

$$P(X) = \sum_{h=0}^{q} u_h \binom{X + q - h}{q - h}, \qquad b_r = \binom{r + q - 1}{q - 1} - P(r) \quad (0 \leq r < m). \tag{9}$$

Binomial expressions in $X$ denote their polynomial expansions.

**Lemma 4.1.** *For every $n \geq 0$, $p_{d,m}(n) = P(n) + b(n)$, where $b$ is the periodic extension of the vector in (9). The leading coefficient of $P$ is*

$$a = \frac{1}{mq!}. \tag{10}$$

*Furthermore,*

$$\Delta^d b(t) = \frac{(-1)^d}{m}(m\mathbf{1}_{\{m \mid t+d\}} - 1). \tag{11}$$

*Proof.* Write $S_m(z) = 1 + z + \cdots + z^{m-1}$. Its expansion at $z = 1$ is

$$S_m(1 - y) = \sum_{k=0}^{m-1} (-1)^k \binom{m}{k+1} y^k.$$

Thus (8) gives the first $q + 1$ coefficients of its reciprocal. The principal part of $(1 - z)^{-q-1}/S_m(z)$ at $z = 1$ is $\sum_{h=0}^{q} u_h (1 - z)^{-q-1+h}$. Its coefficients give $P$ and (10).

At every nontrivial $m$th root of unity $\zeta$, the remaining pole is simple. Multiplication by $1 - z/\zeta$ and passage to the limit gives the partial-fraction term

$$\frac{1}{m(1 - \zeta)^q} \frac{1}{1 - z/\zeta}.$$

These terms account for all poles. Both sides are proper rational functions, so their difference has no polynomial remainder. Hence

$$b(t) = \frac{1}{m} \sum_{\substack{\zeta^m = 1 \\ \zeta \neq 1}} \frac{\zeta^{-t}}{(1 - \zeta)^q}. \tag{12}$$

This is periodic. When $0 \leq r < m$, only $v = 0$ occurs in (7), which verifies the rational formula for $b_r$ in (9).

The cyclic difference acts on $\zeta^{-t}$ by the multiplier

$$2 - \zeta - \zeta^{-1} = -\frac{(1 - \zeta)^2}{\zeta}.$$

Raising this to $d$ cancels the denominator in (12), since $q = 2d$. The sum of all $m$th roots raised to an integer power is $m$ if the power is divisible by $m$, and zero otherwise. Removing the root one proves (11). □

Combining Theorem 2.2 and Lemma 4.1, the coefficient of $n^{2d}$ in the forward transform is

$$\frac{\binom{2d}{d} + 2(-1)^d(m\mathbf{1}_{\{m|n+2d\}} - 1)}{m^2(2d)!}. \tag{13}$$

All lower-order transforms have positive leading coefficients.

4.1. **The four-color example at part size 300.** For the $h_A$ printed in Gajdzica's Example 4.6 [1, Example 4.6], one has $h_A = \mathcal{T}_2(p_{2,300})/2$. With $d = 2$ and $m = 300$, equation (13) therefore gives

$$h_A(n) = \frac{1 + 150\,\mathbf{1}_{\{300|n+4\}}}{1080000}\, n^4 + O(n^3).$$

The exceptional class is $n \equiv 296 \pmod{300}$; 297 is an ordinary class. The least leading coefficient is $1/1080000$, rather than the printed $1/(2^3 \cdot 3^3 \cdot 5^4) = 1/135000$. Thus the printed lower-bound constant is too large for the displayed $h_A$, by a factor of eight on ordinary classes. Both actual coefficients are positive, so the earlier example's conclusion of eventual positivity still holds. The main classification does not depend on these printed numerical details.

## 5. The critical odd case

For even $d \geq 2$, the two numerators in (13) are $\binom{2d}{d} - 2$ and $\binom{2d}{d} + 2(m-1)$, both positive. For odd $d$, the only potentially nonpositive numerator is $\binom{2d}{d} - 2(m-1)$, on $m \mid n + 2d$. This settles every case except

$$d \geq 3 \text{ odd}, \qquad C = \binom{2d}{d}, \qquad m = 1 + C/2. \tag{14}$$

The parameter is an integer because $C = 2\binom{2d-1}{d-1}$.

Put $q = 2d$, $\sigma = (q+m)/2$, and $x = t + \sigma$. The polynomial part has the centered expansion

$$P(t) = ax^q + cx^{q-2} + O(x^{q-4}), \qquad c = -\frac{aq(q-1)(q+m^2)}{24}. \tag{15}$$

Here and below the $O$ notation in polynomial identities means a polynomial of at most the indicated degree; no effective estimate is hidden in it. To verify (15), write $P(t)$ as minus the residue at $z = 1$ of the generating function multiplied by $z^{-t-1}$, and substitute $z = e^{-v}$. Formal residue invariance gives

$$P(t) = \frac{1}{m}[v^q]e^{(t+\sigma)v}\left(\frac{v}{2\sinh(v/2)}\right)^q \frac{mv}{2\sinh(mv/2)}.$$

The product after the exponential is even and equals $1 - (q+m^2)v^2/24 + O(v^4)$, proving (15).

On a residue with $m \mid t + d$, put

$$B_k = \sum_{j=-d}^{d} w_j j^k b(t+j).$$

We require its first three moments:

$$B_0 = -\frac{m-1}{m}, \qquad B_1 = 0, \qquad B_2 = \frac{m-1}{m}\left(-d^2 + \frac{d(2d-1)(m+1)}{6}\right). \tag{16}$$

The first formula follows from (11). For the others, set $S(z) = (2 - z - z^{-1})^d = (-1)^d(1-z)^q z^{-d}$ and $D = z\,d/dz$. Direct differentiation gives

$$\frac{DS}{S} = -d\frac{1+z}{1-z}, \qquad \frac{D^2S}{S} = d^2 + (4d^2 - 2d)\frac{z}{(1-z)^2}. \tag{17}$$

In (12), the critical residue has $\zeta^{-t} = \zeta^d$. The first moment is therefore a constant multiple of $\sum_{\zeta\neq 1}(1+\zeta)/(1-\zeta) = 0$, by pairing inverse roots. The self-inverse root $-1$, if present, contributes zero. For the second moment use

$$\sum_{\substack{\zeta^m=1\\ \zeta\neq 1}} \frac{\zeta}{(1-\zeta)^2} = -\frac{m^2-1}{12}. \tag{18}$$

Indeed, the logarithmic derivative of $S_m$ at 1 gives $\sum(1-\zeta)^{-1} = (m-1)/2$ and $\sum(1-\zeta)^{-2} = (m-1)(5-m)/12$; subtraction proves (18). Equations (12) and (17) now give (16). The operator $D$ acting on $z^{-j}$ introduces $-j$, whereas $D^2$ introduces $j^2$; this sign distinction leaves $B_1 = 0$ unchanged.

**Lemma 5.1.** *Under (14), on the residue class $m \mid t+d$, the residue polynomial of $\mathcal{H}_d(p_{d,m})(t)$ has degree exactly $q-2$, with positive leading coefficient*

$$\beta = \frac{a^2 q! C\, d(m^2-4d^2)}{12} + 2a\binom{q}{2}B_2 + 2cB_0. \tag{19}$$

*Proof.* The degree-$q$ coefficient is zero by (14). There is no degree-$q-1$ coefficient: the smooth part is even in $x$, and the only possible periodic cross term at this degree is a multiple of $B_1 = 0$.

For the next coefficient, expand the moment identity one term further:

$$\sum_j w_j e^{jv} = (-1)^d v^q\left(1 + \frac{d}{12}v^2 + O(v^4)\right).$$

The $a^2$ contribution to $x^{q-2}$ is $-a^2\binom{q}{d+1}(q+2)!d/12$. The mixed smooth contribution is

$$2acq!\left(\binom{q-2}{d} - \binom{q-2}{d-1}\right).$$

It follows either by direct expansion or by the identity

$$(x+j)^q(x-j)^{q-2} + (x+j)^{q-2}(x-j)^q = 2(x^2+j^2)(x^2-j^2)^{q-2}.$$

Substitute (15), $\binom{q}{d+1} = Cd/(d+1)$, and $\binom{q-2}{d-1} = Cd/[2(2d-1)]$. The two contributions reduce to the first term of (19). All lower smooth terms are annihilated at this degree by Lemma 2.1. The cross terms with $b$ contribute the remaining two terms in (19); the purely periodic product has degree zero, which is below $q-2 \geq 4$.

Every displayed summand in (19) is positive. For $d \geq 3$,

$$C \geq \binom{2d}{2} = d(2d-1), \qquad 1 + \frac{d(2d-1)}{2} - 2d = \frac{(2d-1)(d-2)}{2} > 0.$$

Thus $m > 2d$. Also $c < 0$, $B_0 < 0$, and

$$(2d-1)(m+1) > (2d-1)(2d+1) = 4d^2 - 1 > 6d,$$

so $B_2 > 0$. This proves $\beta > 0$. □

*Proof of Theorem 1.1.* The noncritical cases follow from (13), and Lemma 5.1 resolves equality in (3). Every lower order has a positive leading coefficient by Theorem 2.2. Translation from $t$ to $n$ preserves each actual leading sign. Formulas (8)–(9) give the exact input residue polynomials; (6) gives each transform. Taking the maximum of (5) over the finitely many nonzero polynomials

and adjoining 1 gives the asserted $N(d,m)$. Above the cutoff, the negative leading coefficient on $n \equiv -2d \pmod m$ gives the stated repeated failure, with an explicit bound from the same construction. □

*Remark* 5.2. The restriction $d \geq 2$ is essential for this strict classification. For $d = 1, m = 2$, $p_{1,2}(n) = \lfloor (n+2)^2/4 \rfloor$ and $\mathcal{T}_1(p_{1,2})(2k) = 0$, $\mathcal{T}_1(p_{1,2})(2k+1) = 2(k+2)^2 > 0$. For $m \geq 3$, (13) has a negative residue.

## 6. Computational and formal verification

The mathematical proofs above are algebraic and do not depend on finite experiments. The accompanying code generates rational residue polynomials, their transformed coefficient vectors, and the explicit sign bounds. The initial certificate set includes the critical pairs $(3, 11)$ and $(5, 127)$, the pairs immediately above and below the latter cutoff, and the previously known pair $(2, 300)$. Two separate model sessions reviewed the paper argument; their single minor threshold-convention issue was repaired.

Independent Newton interpolation from direct partition counts, followed by recurrence and initial-value checks, reproduces all coefficients, signs, and bounds in 724 partition records and 14 boundary test polynomials. A counterexample search covered 63 parameter pairs (all orders up to the specified order) and 56 rational periodic examples with a fixed seed. All 41,705 exact identity checks passed; point and bound checks are included in that total. These finite checks do not prove the infinite-parameter theorem.

The Lean-verified component is the algebraic cancellation

$$\frac{(2 - z - z^{-1})^d}{(1-z)^{2d}} = \frac{(-1)^d}{z^d} \qquad (z \in \mathbb{C} \setminus \{0, 1\},\ d \in \mathbb{N}),$$

used in (11). Further checked statements establish the sign criterion for the explicit numerators in (13), positivity of the expression (19), the transformation formula (6) for a supplied rational residue representation, and the rational version of the coefficient bound (5), including a common finite bound. These components pass Lake builds with Lean 4.35.0 and Mathlib and independent blind statement comparisons. The axiom audits report only `propext`, `Classical.choice`, and `Quot.sound`, without admitted proofs, custom axioms, or `native_decide`. The original targets now have author-supplied unconditional Lean proofs: `main_classification_proved` and `above_threshold_failure_proved`. The concrete Bernoulli residue construction is for the same finite partition sum; not every displayed partial-fraction identity has been formalized verbatim. Author checks report a clean project rebuild, axiom audits and `leanchecker -fresh`. Independent statement and formal audits have passed for the exact two targets in the fixed version `F-fed30c64910e057406f3342025deff71`. The corrected edition preserves the accepted proof sources. A companion Lean regression specializes the accepted boundary coefficient to the normalization of Example 4.6; a separate exact arithmetic script verifies all 300 residue polynomials from the partition counts. These checks do not extend the scope of formal acceptance. The general real-coefficient bound is not claimed formalized.

## 7. Concluding remarks

The centered polynomial and two Fourier moments resolve the vanishing boundary coefficient for (2). Finding similarly explicit critical coefficients for other finite multisets is a natural further question.

The documented search found no equivalent full classification. Access to some databases and a related final Acta Arithmetica article was incomplete; global publication absence is not proved. The uninspected article is not used in the argument, and the known smooth criterion and earlier example are explicitly credited above.

## Use of AI tools

OpenAI Codex language models assisted the original formulation, proof development, computation and exposition. For this corrected edition, a GPT-6 Astra subagent at Max reasoning independently rederived the specialization to Example 4.6. GPT Astra appears in the byline as an AI contributor. This credit is not a claim of human authorship or journal peer review.

## Data and code availability

The corrected paper, exact arithmetic script, residue coefficients, Lean regression and versioned verification records are available in https://github.com/1200r3/math-research-results/tree/main/projects/qp01. The original publication and fixed mathematical record remain available in the repository history. The paper and original prose are CC BY 4.0; the code and Lean sources are MIT-licensed.

## References

[1] Krystian Gajdzica. The turán and laguerre inequalities for quasi-polynomial-like functions. *Research in Number Theory*, 10(2), April 2024. URL: http://dx.doi.org/10.1007/s40993-024-00532-7, doi:10.1007/s40993-024-00532-7.